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Action spectrum of the redox state of the plastoquinone pool defines its function in plant acclimation

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SUMMARY

The plastoquinone (PQ) pool mediates electron flow and regulates photoacclimation in plants. Here we report the action spectrum of the redox state of the PQ pool in *Arabidopsis thaliana*, showing that 470–500, 560 or 650–660 nm light favors Photosystem II (PSII) and reduces the PQ pool, whereas 420–440, 520 or 690 nm light favors Photosystem I (PSI) and oxidizes PQ. These data were used to construct a model predicting the redox state of PQ from the spectrum of any polychromatic light source. Moderate reduction of the PQ pool induced transition to light state 2, whereas state 1 required highly oxidized PQ. In low-intensity PSI light, PQ was more oxidized than in darkness and became gradually reduced with light intensity, while weak PSII light strongly reduced PQ. Natural sunlight was found to favor PSI, which enables plants to use the redox state of the PQ pool as a measure of light intensity.

Keywords: plastoquinone, high-performance liquid chromatography, F_0 rise, post-illumination fluorescence rise, action spectroscopy, state transitions, sunlight, white light, plastocyanin, *Arabidopsis thaliana*.

INTRODUCTION

Two protein complexes, Photosystem II (PSII) and Photosystem I (PSI), absorb light in photosynthesis. The photochemically active plastoquinone (PQ) pool, containing 3.6–6.0 PQ molecules per one PSII, functions as an electron carrier between PSII and PSI (Graan and Ort, 1984; McCauley and Melis, 1986; Kruk and Karpinski, 2006). PSII reduces photochemically active PQ to PQH_2, which is oxidized back to PQ by the cytochrome b_6/f complex (Cyt b_6f). One electron is transferred from Cyt b_6f to the soluble electron carrier plastocyanin (PC) that reduces the primary donor of PSI, P_{700}^+. The second electron returns to PQ via the internal electron transfer chain of Cyt b_6f (Tikhonov, 2014). In addition to the photochemically active pool (hereafter the PQ pool) that can be reduced or oxidized by light, PQ and plastoquinol (PQH_2) are also found in plastoglobuli and in the chloroplast envelope membrane (Kruk and Karpinski, 2006).

In moderate light, the redox state of the PQ pool depends on the ratio of electron transfer rates (ETRs) of the two photosystems, as preferential excitation of PSII or PSI causes reduction or oxidation of the PQ pool, respectively. Strong light, in turn, reduces the PQ pool because the maximum rate of oxidation of PQH_2 by Cyt b_6f is slower than the delivery of electrons to the PQ pool by PSII (Hope *et al*., 1988; Laisk *et al*., 2005). In the dark, the PQ pool is partially reduced by the mechanisms of cyclic electron flow, with ferredoxin and possibly also NADPH as ultimate electron donors (Mills *et al*., 1979; Shikanai *et al*., 1998) and partially oxidized by the plastid terminal oxidase (Lennon *et al*., 2003).

The PQ pool is a key regulator of plant acclimation (Rochaix, 2014). Firstly, the redox state of the PQ pool regulates state transitions, which are short-time compensatory acclimation reactions of chloroplasts (Allen *et al*., 1981; Zito *et al*., 1999; Pesaresi *et al*., 2002; Chuartzman *et al*., 2008; Nawrocki *et al*., 2016; Koskela *et al*., 2018). Light favoring PSI induces State 1 in which the light-harvesting complex II (LHCII) serves PSII, whereas light favoring PSII induces State 2, in which the LHCB1 and LHCB2 proteins become phosphorylated and some LHCII trimers serve PSI (Zito *et al*., 1999; Bellafiore *et al*., 2005; Pietrzykowska *et al*., 2014). Secondly, the redox state of the PQ pool triggers retrograde signal(s) that affect the expression of nuclear genes (Escoubas *et al*., 1995; Karpinski *et al*., 1997; Pfannschmidt *et al*., 2001; Piippo *et al*., 2006; Adamiec *et al*., 2008; Dietzel *et al*., 2015). This regulation functions, at least partly, via alternative splicing (Petrillo *et al*., 2014).

Thirdly, the expression of several chloroplast genes depends on the redox state of the PQ pool (Pfannschmidt *et al.*, 1999; Tullberg *et al.*, 2000; Schütze *et al.*, 2008). Long-time light acclimation (Wagner *et al.*, 2008; Dietzel *et al.*, 2015) has been shown to depend on the redox state of the PQ pool (Wagner *et al.*, 2008; Bode *et al.*, 2016). Furthermore, PQ plays a role in temperature responses (Szechyńska-Hebda *et al.*, 2010) and may function as an antioxidant (Kruk and Szymańska, 2012; Ksas *et al.*, 2015; Khorobrykh and Tyystjärvi, 2018).

Studies of state transitions, LHCII phosphorylation, gene regulation and long-time acclimation to light quality have so far focused on extreme states 1 and 2. The involvement of the PQ pool has often been tested with 3-(3,4-dichlorophenyl)-1,1-dimethylurea (DCMU) that blocks PSII, causing oxidation of the PQ pool in the light, and/or 2,5-dibromo-6-isopropyl-3-methyl-1,4-benzoquinone (DBMIB) that blocks the quinone oxidation site in Cyt b_6f, causing reduction of the PQ pool in the light (Allen *et al.*, 1981; Karpinski *et al.*, 1997; Piippo *et al.*, 2006; Szechyńska-Hebda *et al.*, 2010; Petrillo *et al.*, 2014; Bode *et al.*, 2016). In the absence of these inhibitors, the wavelength distribution of ambient light, the wavelength preferences of the photosystems and the light state of the leaf determine the redox state of the PQ pool. Photoacoustic measurements (Canaani and Malkin, 1984; Veeranjaneyulu and Leblanc, 1994) and *in vitro* absorption spectra (Wientjes *et al.*, 2013) suggest that in State 2, some wavelengths of visible light favor PSII, some PSI (Canaani and Malkin, 1984; Veeranjaneyulu and Leblanc, 1994; Wientjes *et al.*, 2013), but in State 1 PSII has a higher quantum yield than PSI (Canaani and Malkin, 1984; Veeranjaneyulu and Leblanc, 1994). A higher quantum yield for PSII was also measured when the light state was not controlled (Laisk *et al.*, 2014). Thus, the effects of wavelengths on the redox state of the PQ pool may depend on whether plants are nearer to State 1 or State 2 in growth light. However, no action spectrum of the redox state of the PQ pool has been measured.

The present study aims at filling the knowledge gap between the well-known importance of the redox state of the PQ pool and the lack of knowledge on how the PQ pool responds to quality and quantity of photosynthetically active light. We developed a chlorophyll (Chl) *a* fluorescence method, activated F_0 rise, to screen for wavelengths favoring PSII or PSI, and then applied a high-performance liquid chromatography (HPLC) method (Kruk and Karpinski, 2006) for actual measurements of the redox state of the PQ pool. The measurements revealed that several wavelengths of photosynthetically active light cause strong reduction or oxidation of the PQ pool. Various broadband light sources caused intermediate redox states and, based on the HPLC data, a model was developed for the estimation of the redox state of the PQ pool under various light sources. The light intensity response of the PQ pool under PSI light, PSII light and sunlight showed that natural sunlight favors PSI. The results reveal a curvilinear relationship between the photosynthetic light state and the redox state of the PQ pool, and show that in sunlight the PQ pool can also measure light intensity.

RESULTS AND DISCUSSION

An F_0 rise method for probing wavelengths favoring PSII or PSI

A transient post-illumination increase in Chl *a* fluorescence yield, the F_0 rise, is observed when illumination of an intact leaf ends (Figure 1a). At the end of the illumination, electron transfer through PSII and PSI ceases but reduction of PQ by stromal reductants via the routes of cyclic electron flow continues (Mills *et al.*, 1979; Endo *et al.*, 1997), causing a transient increase in Chl *a* fluorescence yield (Groom *et al.*, 1993; Shikanai *et al.*, 1998; Rumeau *et al.*, 2005). An abnormally low F_0 rise can reveal a defect in the molecular machinery of cyclic electron flow (Shikanai *et al.*, 1998; Rumeau *et al.*, 2005).

Traditionally, the measuring beam (MB) of the pulse amplitude modulation fluorometer has been continuously on during the F_0 rise measurements. We noticed that a chopped MB (0.6 sec on/5 sec off) does not induce F_0 rise (Figure 1b). F_0 rise was restored when weak activating light was applied on top of the chopped MB (Figure 1). This activated F_0 rise was absent in the *ndh-0* mutant lacking cyclic electron flow that depends on the thylakoid NDH complex that mediates electron flow from ferredoxin to the PQ pool (Shikanai *et al.*, 1998; Rumeau *et al.*, 2005) and high in the *high cyclic electron flow* (*hcef*) mutant (Livingston *et al.*, 2010), indicating that the molecular machinery of cyclic electron flow is a prerequisite for activated F_0 rise (Figure 1). However, the requirement of the activating light indicates that PQH_2 accumulation by the mechanisms of cyclic electron flow does not directly reduce Q_A, but instead the MB (in traditional F_0 rise measurement) or the activating light (in activated F_0 rise) reduces Q_A, producing F_0 rise. In the physiological context, the result indicates that F_0 rise actually is variable fluorescence. PQH_2 accumulation increases the occupation of the Q_B site where PQH_2 inhibits reoxidation of Q_A^- (De Wijn and van Gorkom, 2001; Oja *et al.*, 2011). Thus, activated F_0 rise can be used to probe whether the activating light favors PSII or PSI, as PSII light efficiently reduces Q_A and thereby promotes a high F_0 rise, while PSI light suppresses F_0 rise by oxidizing the PQ pool.

Several factors, for example activity of the NDH complex (Figure 1), conditions of the stroma and non-photochemical quenching (NPQ) induced by the pre-illumination, may affect the magnitude of the F_0 rise. Therefore, use of F_0 rise as a quantitative method requires accurate adjustment of the photon flux density (PFD) of the activating light,

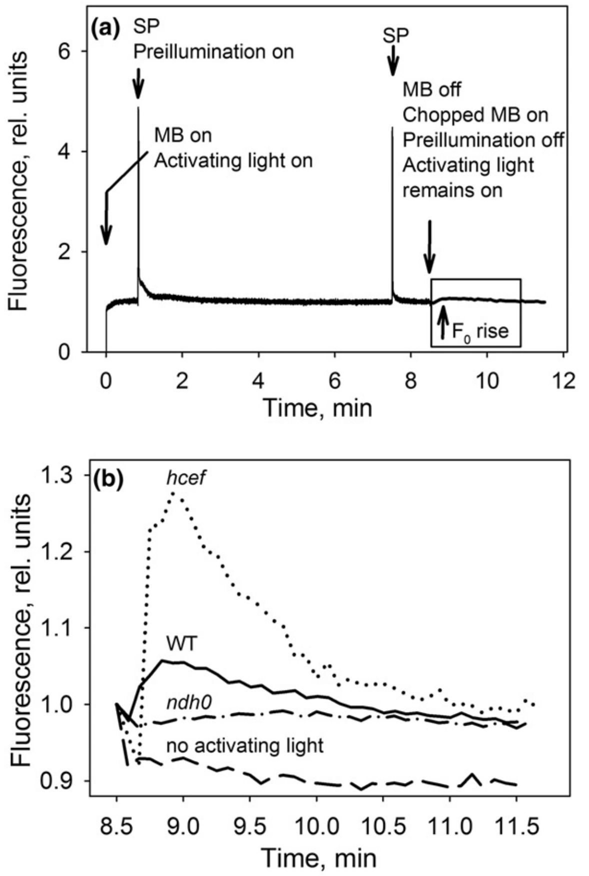

Figure 1. Measurement of activated F_0 rise, and results from wild-type (WT) and mutant *Arabidopsis thaliana*.
(a) Measurement protocol of activated F_0 rise. After pre-illumination with a halogen lamp, at the photosynthetic photon flux density (PPFD) of 47.5 μmol m^{-2} sec^{-1}, applied together with weak activating light (PPFD 2.5 μmol m^{-2} sec^{-1}), the pre-illumination was switched off but the activating light was continuously applied on top of a chopped measuring beam (MB; 0.6 sec on/5 sec off). SP indicates a saturating pulse. The position of the F_0 rise is indicated with a mark and a box.
(b) Enlargement of the boxed area showing activated F_0 rise in WT *A. thaliana* (solid and dashed line), and result from *ndh-0* (dash-dotted line) and *hcef* (dotted line), obtained using white light as pre-illumination and 660-nm light as activating light (solid, dotted and dash-dotted line, respectively). The dashed line shows a measurement without activating light. Each curve in (b) represents an average of measurements from three leaves taken from different plants.

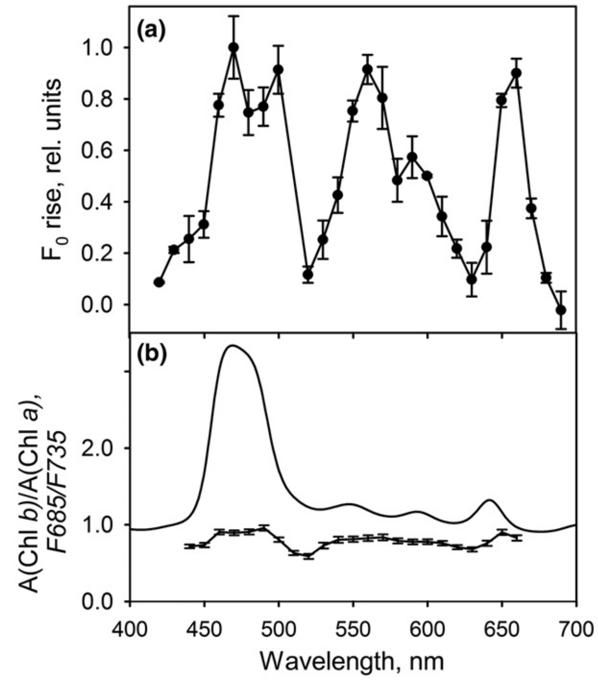

Figure 2. Action spectrum of activated F_0 rise, chlorophyll (Chl) *a/b* absorbance ratio and Photosystem II (PSII)/Photosystem I (PSI) fluorescence ratio.
(a) Action spectrum of activated F_0 rise in *Nicotiana tabacum* leaves.
(b) Ratio of Chl *b* to Chl *a* absorbance in methanol (solid line), calculated from earlier data (Frigaard *et al*., 1996; obtained in a table form from the author) and excitation spectrum of the ratio of Chl *a* fluorescence at 685 nm (PSII fluorescence) to 735 nm (PSI fluorescence), measured by illuminating isolated thylakoids at different wavelengths at 77 K (line with error bars). Each experimental point represents an average of pieces cut from three–four different *N. tabacum* leaves, those in (b) represent measurements from three batches of *Arabidopsis thaliana* thylakoids, each isolated from a different plant, and the error bars show SEM.

standardized pre-illumination and careful consideration of the illumination history of the plant before the experiment.

Action spectrum of activated F_0 rise identifies wavelengths favoring PSII or PSI

An action spectrum of the activated F_0 rise was measured at 10 nm resolution from *Nicotiana tabacum* leaves to find wavelengths of the maxima and minima of the PSII/PSI absorbance ratio. A high F_0 rise was obtained with blue 460–500 nm, green 560–570 nm and red 650–660 nm activating light (Figure 2), indicating that these wavelengths favor PSII. Blue 420–440 nm, green 520 nm, red 630 nm and far-red 680–690 nm activating light suppressed F_0 rise (Figure 2a), indicating that these wavelengths are better absorbed by PSI than by PSII. Wavelengths between the peaks and minima showed intermediate F_0 rise values. The action spectrum suggests PSII absorbs more light than PSI at the peaks of the spectrum, and PSI absorbs more than PSII at the minima, but the data do not tell how big the absorbance differences are. Furthermore, the action spectrum of F_0 rise only reveals the ratio of the absorption flux of PSII to that of PSI – for example, at 560 nm PSII absorbs more than PSI although the absorbance of both photosystems is relatively weak, as Chls *a* and *b* absorb little at this wavelength.

To measure the absorbance differences causing the wavelength dependence of F_0 rise, we measured the ratio of 685 nm PSII fluorescence to 735 nm PSI fluorescence from *Arabidopsis thaliana* thylakoids at 77 K. The results (Figure 2b) showed that the ratio of PSII fluorescence to PSI fluorescence has peaks and minima at the same excitation wavelengths as F_0 rise, although the dynamics of the fluorescence ratio are modest, as the highest peak is only 1.6 times as high as the minimum. The finding that the absorbance properties of the two photosystems have only minor differences suggests that polychromatic light may favor either PSII or PSI, and that small state transitions can compensate for a changed balance between PSII and PSI upon a change in the wavelength distribution.

Absorbance differences between the two photosystems depend on their pigment compositions, and the main difference between the photosystems is the higher amount of Chl *b* in PSII than in PSI (Wei *et al*., 2016; Caspy and Nelson, 2018). Figure 2(b) shows that the absorbance peaks of Chl *b* may explain why 460–500 nm and 650–660 nm favor PSII, whereas the PSI preference of 420–440 nm light may reflect the strong absorption by Chl *a* in this region. The green PSII light region at 560–570 nm is not located at a wavelength range of strong absorption by Chl *b*, and 520 and 630 nm regions that favor PSI do not coincide with strong peaks of Chl *a* absorption. Still, comparison of the action spectrum of F_0 rise with the ratio of the absorbance spectra of Chls *b* and *a* in methanol (Frigaard *et al*., 1996; Figure 2b) suggests that the photosystem preferences of these wavelengths can be traced to the absorbance profiles of Chls *b* and *a*. The two spectra resemble each other, although the Chl *b/a* absorbance ratio is high in blue light and varies with excitation wavelength only little in other wavelengths. In the 548–676 nm range, the local peaks and minima of the Chl *b/a* absorbance ratio are found at approximately 10 nm shorter wavelengths than the respective extremes of the action spectrum of F_0 rise, probably because the absorption peaks of both Chls are found at longer wavelengths in photosynthetic proteins than in organic solvents (Moss and Loomis, 1952; Thomas and van der Wal, 1964). The matching positions of the peaks and minima of the F685/F735 ratio from *A. thaliana* (Figure 2b) with the peaks and minima of the action spectrum of F_0 rise from *N. tabacum* (Figure 2a) confirm that the wavelengths favoring PSII or PSI are independent of plant species.

Amount of PQ and size of the photochemically active PQ pool

The connection between F_0 rise and the redox state of the PQ pool promoted us to study if the redox state of the PQ pool has an action spectrum resembling that of F_0 rise. The amount of total PQ (i.e. PQ + PQH_2) per leaf area was quantified by measuring the PQ/Chl ratio with HPLC, and comparing with spectrophotometrically determined amounts of Chl *a* and *b* per leaf area. The molar ratio of PQ to Chl was 0.0193 $\pm$ 0.0013, and the amount of Chl (*a* + *b*) was 255.4 μmol m^{-2}, indicating that *A. thaliana* leaves contained 4.93 μmol PQ m^{-2}. Comparison of the ratio of the complementary area (CA) above fluorescence induction curves (Vernotte *et al*., 1979) measured in the absence and presence of DCMU suggested that *A. thaliana* leaves contain 6.29 $\pm$ 1.05 photochemically active PQ per PSII (Figure S1).

For the measurements of the redox state of the PQ pool, PQ was extracted by shortly grinding the illuminated leaf samples in dry cold ethyl acetate (Kruk and Karpinski, 2006). Rapid dark-induced oxidation of PQH_2 in leaf fragments (Hope *et al*., 1988; Laisk *et al*., 2005) was avoided by continuing illumination throughout the grinding, and a separate test with a transparent mortar confirmed that the reduction state did not change during grinding (Table S1). Once in organic solvent, PQH_2 is oxidized only very slowly (Kruk and Strzałka, 1999; Khorobrykh and Tyystjärvi, 2018). Each sample was divided into two parts, one assayed directly and the second one after addition of sodium borohydride that reduces PQ to PQH_2 (Kruk and Karpinski, 2006); the remaining PQ peak in the reduced samples represented less than 10% of total PQ (Figure S2). The size of the photochemically active PQ pool was determined by measuring the ratio PQH_2/(PQ + PQH_2) from leaf samples illuminated with far-red light to oxidize the photochemically active pool and from samples illuminated with strong light to reduce the pool. These measurements showed that 48.9 $\pm$ 7.9% of PQ, or 2.41 μmol PQ m^{-2}, belonged to the photochemically active pool. Similar results have been obtained by freezing the leaf samples in liquid nitrogen prior to the extraction in the dark (Ksas *et al*., 2015), and a somewhat smaller photochemically active fraction (30–35%) has been found in earlier HPLC measurements done without freezing (Kruk and Karpinski, 2006; Szechyńska-Hebda *et al*., 2010; Yoshida *et al*., 2010). Also in cyanobacteria, only part of total PQ belongs to the photochemically active pool (Khorobrykh *et al*., 2020).

In dark-acclimated leaves, 26% of photochemically active PQ was found to be reduced (Figure 3), in line with earlier results (Kruk and Karpinski, 2006; Szechyńska-Hebda *et al*., 2010; Yoshida *et al*., 2010). Exchange of PQ between the thylakoid pool and plastoglobuli during the 30-sec illumination time was negligible, as control experiments yielded 80.23 $\pm$ 2.61% of PQH_2 in total PQ + PQH_2 with 5-sec high-light illumination after a 2-h dark period, and 81.93 $\pm$ 4.02% after 30-sec illumination.

Wavelength dependence of the redox state of the PQ pool in *Arabidopsis thaliana*

Next, an action spectrum of the redox state of the photochemically active PQ pool was measured from light-acclimated *A. thaliana* leaves using wavelengths showing the strongest PSII or PSI preferences. Illumination at the blue, green or red PSII light at 470, 560 or 660 nm reduced the PQ pool by 78–92%, whereas 440, 520 or 690 nm PSI light oxidized more than 90% of the PQ pool (Figure 3). Orange light of 630 nm that acted as PSI light when applied at very

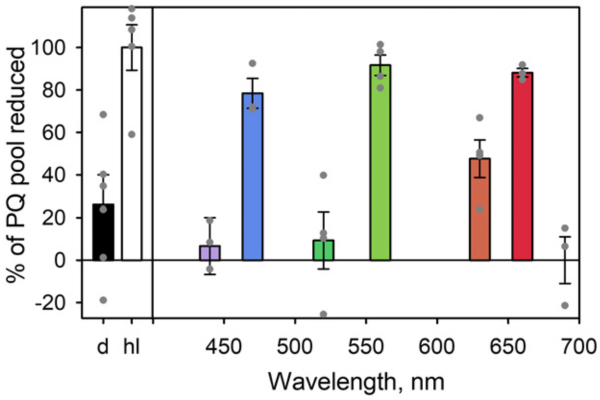

Figure 3. Action spectrum of the redox state of the photochemically active photoquinone (PQ) pool in *Arabidopsis thaliana* leaves.
The redox state of the PQ pool was measured with high-performance liquid chromatography (HPLC) after 2-h dark incubation (black bar), after a 30-sec high light pulse after 2-h dark incubation (white bar), and after 4 min illumination of light-acclimated leaves at photosynthetic photon flux density (PPFD) 50 μmol m^{-2} sec^{-1} with monochromatic light (colored bars; wavelength on *x*-axis). Each measurement represents an average of measurements from three *A. thaliana* leaves, each from a different plant, and the bars show SEM. The gray dots show the individual data points.

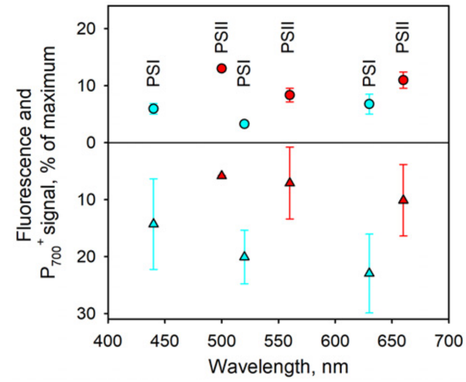

Figure 4. Action spectra of steady-state fluorescence yield and P_{700}^{+} signal.
Chlorophyll (Chl) *a* fluorescence yield representing Photosystem II (PSII; circles) and P_{700}^{+} signal of Photosystem I (PSI; triangles) after 2-min illumination of light-adapted *Arabidopsis thaliana* at photosynthetic photon flux density (PPFD) 50 μmol m^{-2} sec^{-1} with different wavelengths. Maximum fluorescence was measured during a pulse of saturating white light, and the maximum of P_{700}^{+} signal was obtained with a far-red pulse. The labels 'PSI' and 'PSII' indicate which photosystem is favored according to the action spectrum of F_0 rise (Figure 2). Each symbol represents an average of measurements from three different leaves, and the error bars show SD.

low intensity in F_0 rise experiments (Figure 2) caused an intermediate redox state at the moderate intensity used in PQ measurements (Figure 3). These results indicate that the photosystem preferences of the wavelengths exert large effects on the redox state of the PQ pool under monochromatic light of moderate intensity, although the absorption differences between the photosystems are modest (Figure 2); see also Canaani and Malkin (1984), Veeranjaneyulu and Leblanc (1994), Wientjes *et al*. (2013), Laisk *et al*. (2014), and Hogewoning *et al*. (2012).

Photosynthetic electron transfer reflects the state of the PQ pool

A reduced PQ pool promotes high Chl *a* fluorescence by slowing down reoxidation of Q_A (De Wijn and van Gorkom, 2001; Oja *et al*., 2011) and, indeed, leaves illuminated with wavelengths favoring PSII had a higher Chl *a* fluorescence yield than leaves illuminated with PSI wavelengths (Figure 4). Accordingly, the combined absorbance signal of the oxidized primary donor of PSI (P_{700}^{+}) and PC (Schreiber *et al*., 1988) was weaker in PSII than in PSI light (Figure 4). However, both the steady-state Chl *a* fluorescence yield and the P_{700}^{+} signal were far from the respective maxima (Figure 4), as expected in moderate light.

The quantum yield of PSII electron transfer (Φ_{II}) was relatively high throughout the visible spectrum, varying between 0.61 and 0.78 (Figure 5). The PSII ETR was calculated by taking into account the wavelength dependences of Φ_{II} and leaf absorptance (Figure 5). Leaf absorptance peaked at 482–492 and 675–682 nm, in agreement with earlier measurements (Moss and Loomis, 1952). The PSI/PSII ratio was assumed to be 1.06 (Yin *et al*., 2012). The data show that ETR varied between 16.3 (at 680 nm) and 10.3 (at 550 nm) μmol electrons m^{-2} sec^{-1} (Figure 5). The minima of Φ_{II} and ETR values occurred at the PSII wavelengths where the PQ pool was reduced, but these minima were only 15–36% lower than the average values of these parameters, indicating that photosynthetic electron transfer functioned perfectly even if the PQ pool was highly reduced under monochromatic PSII light. The fluorescence parameter 1-qL (Kramer *et al*., 2004) that quantifies the fraction of closed PSII centers stayed between 0.52 and 0.98 at all wavelengths (Figure 5), and the Pearson correlation coefficient between 1-qL and the redox state of the PQ pool was only 0.74 (Figure S3). Thus, photochemical quenching correlates with the redox state of the PQ pool but fails as a quantitative measure in our experimental conditions, indicating that actual PQ measurements are superior to photochemical quenching assays for the elucidation of the redox state of the PQ pool.

Plastocyanin is reduced by Cyt b_6f and oxidized by P_{700}, analogously to PQ that is reduced by PSII and oxidized by Cyt b_6f. The redox state of PC is therefore expected to respond to imbalance in excitation between PSII and PSI in the same way as PQ. We measured the wavelength effects on the redox state of PC with the Dual/KLAS-NIR spectrophotometer (Schreiber and Klughammer, 2016) after minor modifications that allowed us to illuminate the sample with specific wavelengths during the measurement. The results (Figure 6) show that both P_{700} and PC became oxidized under 440, 520, 630 and 690 nm light, and stayed

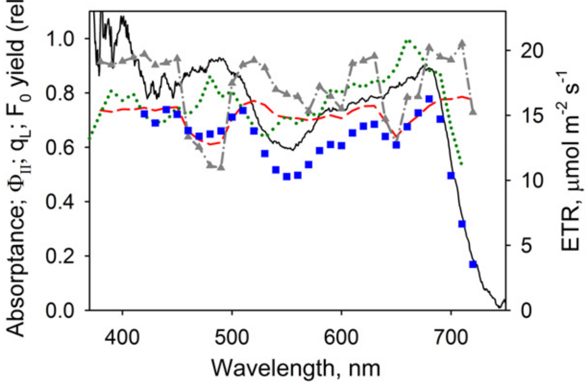

Figure 5. Action spectra of photosynthetic electron flow and fluorescence in *Arabidopsis thaliana* leaves.
Leaf absorptance (black solid line), action spectrum of the quantum yield of Photosystem II (PSII) electron transfer (Φ_{II}) (red dashed line), action spectrum of qL (gray triangles, dash-dotted line), action spectrum of F_0 (green dotted line) and action spectrum of photosynthetic electron transfer rate (ETR; blue squares). ETR was calculated as photosynthetic photon flux density (PPFD) × Φ_{II} × leaf absorptance × 0.485, where the constant factor was calculated from the data in Yin *et al.* (2012). Φ_{II}, qL and ETR were measured at PPFD 50 μmol m^{-2} sec^{-1}, and weak, short light pulses were used to measure F_0. Each data point of Φ_{II}, qL and F_0 represents an average of three different leaves; leaf absorptance was measured from one batch of 10 leaves.

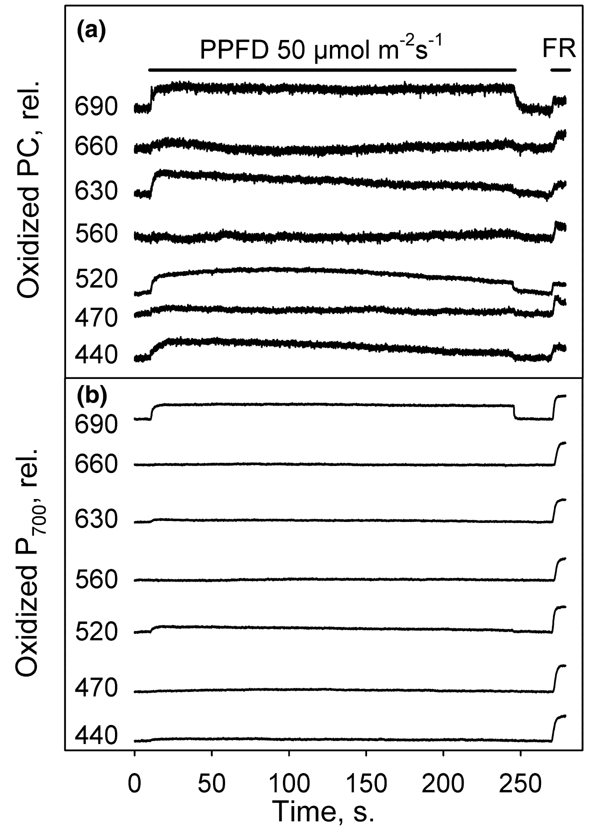

Figure 6. Effects of selected wavelengths on plastocyanin (PC) and P_{700}.
Redox changes were measured with Dual/KLAS-NIR spectrometer. For each experiment, a light-acclimated *Arabidopsis thaliana* leaf was darkened for 1 min, after which the measurement of PC and P_{700} redox changes started. Illumination with monochromatic light was then started after 10 sec, and continued for 240 sec. Finally, a 10-sec far-red pulse was applied after 30 sec of darkness.
(a) PC redox kinetics. (b) P_{700} redox kinetics. The curves have been shifted in the *y*-axis direction to facilitate comparison, and the numbers indicate the wavelength of the illumination in nm. Each curve represents a single, representative measurement out of measurements from three leaves, each from a different plant.

fully reduced at 560 nm and almost fully reduced at 470 and 660 nm light. The initial oxidation of PC at 440, 520 and 630 nm light diminished during the 4-min illumination. The effects of 440, 520 and 630 nm light on the level of oxidation of P_{700} were small, and even 690 nm light caused less P_{700} oxidation than the intense far-red light of the instrument (Figure 6). At the beginning of the illumination, all PSI wavelengths oxidized PC more than the far-red treatment at the end of the measurement. These data show that the redox state of PC follows that of PQ, especially at the beginning of illumination after a short dark period.

The kinetics of reduction of PC upon an abrupt stop of illumination indicate the availability of electrons in the PQ pool at the end of illumination. Measurements with the Dual/KLAS-NIR spectrophotometer showed that the reduction of PC after 4-min illumination with 690 or 520 nm light takes several seconds, strongly suggesting that the electrons necessary for the reduction were not present in the PQ pool at the end of illumination (Figure S4), and thus confirming that the PQ pool was highly oxidized at the end of the illumination. After 440 and 630 nm illumination, PC was reduced more rapidly but the amplitude was very small, and after 470, 560 or 660 nm light, no further reduction of PC occurred (Figure S4). Re-reduction of P_{700} occurred at a much faster rate than re-reduction of PC, but the amplitude was significant only after illumination with 690 nm light (Figure S4). These results confirm the data (Figures 2b and 4) showing that the absorbance differences of the photosystems between visible light wavelengths are small.

Spectral distribution of polychromatic light determines its effect on the PQ pool

Wavelengths determine the redox state of the PQ pool via absorbance differences of the photosystems, suggesting that the effect of polychromatic light can be calculated by summing the effects of the constituent wavelengths. To test this possibility, the redox state of the PQ pool was measured under moderate-intensity illumination with seven different white light sources. Furthermore, 13 monochromatic sources from ultraviolet-A to far-red and with varying spectral widths were tested. The monochromatic sources revealed that wavelengths shorter than

440 nm, including the tested ultraviolet-A wavelengths, favor PSI, and the effect of far-red light at 721 nm was similar to that of 690 nm light (Figure 7a). Wideband light, 36 nm full-width at half-maximum (FWHM), centering at 554 nm caused much less reduction of the PQ pool than a narrow band (10 nm FWHM) at 560 nm (Figure 7a). The redox state of the PQ pool varied between the white light sources, but was less than 50% reduced in all measured types of white light (Figure 7a). The PQ pool was fairly oxidized under incandescent lamp or the KL-1500 halogen illuminator that have a large far-red contribution in the spectrum, but a warm white LED or an energy saver lamp that emit very little far-red induced only marginally higher reduction of the PQ pool. A high-pressure sodium lamp induced the highest reduction level of the white light sources, as 42% of the PQ pool was reduced (Figure 7a).

To better understand the effect of polychromatic light, we constructed and optimized a model for calculating the redox state of the PQ pool from the spectrum of illumination. For each spectrum, the contribution of each wavelength (at 1 nm steps) to the total photosynthetic photon flux density (PPFD) of 50 μmol m^{-2} sec^{-1} was calculated, and each wavelength contribution was multiplied with an optimized coefficient. The coefficients ranged from −50 to +50 for fully oxidized to fully reduced PQ pool, and an overshoot of ± 5 was necessary to give flexibility for the model. The sum of the products (coefficient times PFD at each wavelength) was calculated to obtain the predicted redox state of the PQ pool. A Monte Carlo procedure was used to optimize the initially random coefficients so that the results matched with experimental data. The coefficients (Figure 7b; see Supporting Information for the application of the model) resemble the action spectrum of F_0 rise (Figure 2a). The predicted redox state of the PQ pool, obtained by multiplying the spectrum with the coefficients, agreed well with experimental data (Figure 7a).

Next, we analyzed the polychromatic spectra using the wavelength coefficients to study the spectral features that cause the photosystem preferences of the spectra. The spectrally resolved effects of the polychromatic light sources (Figure 8) reveal that green and orange light at 505–540 and 595–630 nm, respectively, may contribute more to oxidation of the PQ pool than far-red light (> 700 nm). Moreover, the strong PQ-oxidizing effect of far-red light already starts at 680 nm. Red and green wavelengths favoring PSII (peaks at 662 and 563 nm, respectively) have in most light sources equally high importance for the reduction of the PQ pool. Blue light contributes significantly to the redox state of the PQ pool in a cool white LED (Figure 8e).

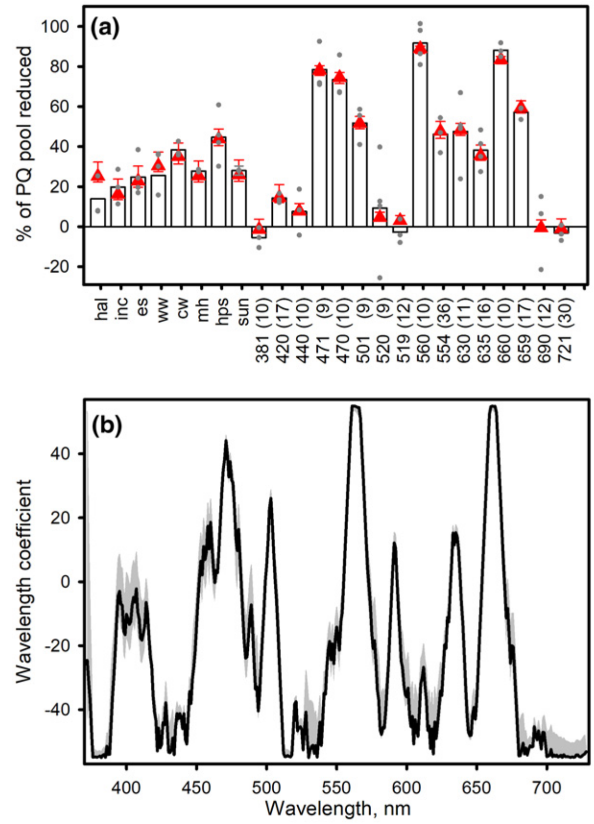

Figure 7. Effects of different light sources on the redox state of the plastoquinine (PQ) pool.
(a) Measured (bars) and modeled (triangles) redox state of PQ pool after 4-min illumination of light-acclimated leaves at photosynthetic photon flux density (PPFD) 50 μmol m^{-2} sec^{-1} with KL-1500 halogen (hal), incandescent (inc) and energy saver (es) lamp, warm white (ww) and cool white (cw) LED, multimetal halide (mh) and high-pressure sodium (hps) lamp, sunlight (sun) and in monochromatic light shown with wavelengths in nm, as indicated. The full-width at half-maximum (FWHM), in nm, of the spectral peak of each monochromatic source is shown in parentheses. Each bar represents an average of measurements from three leaves, each from a different plant; the value and error estimate for sunlight were obtained by interpolation from data obtained at PPFD 30 and 106 μmol m^{-2} sec^{-1}. The error bars of the modeled data show the results of a sensitivity analysis for 10% increase in χ^2. The gray dots show the results of individual experiments. The experimental results at 440, 470, 520, 560, 630, 660 and 690 nm are from Figure 3.
(b) Wavelength coefficients, positive for reduction and negative for oxidation of the PQ pool. The gray area shows the sensitivity of the coefficient value for 10% increase or decrease in the predicted percentage of PQH_2.

Sunlight is PSI light

The PQ pool was found to respond in profoundly different manners to different intensities of monochromatic PSII and PSI light, as PSII light reduced the PQ pool already at low intensity, whereas increasing intensity of PSI light caused a gradual shift from oxidized to reduced PQ (Figure 9). The intensity response of the PQ pool under natural sunlight resembled that of PSI light, as low-intensity sunlight oxidized the PQ pool below the dark level (Figure 9), 50% reduction was reached above PPFD 100 μmol m^{-2} sec^{-1}, and 70–80% of the PQ pool was found to be reduced in intense sunlight (Figure 9a). Accordingly, the reduction

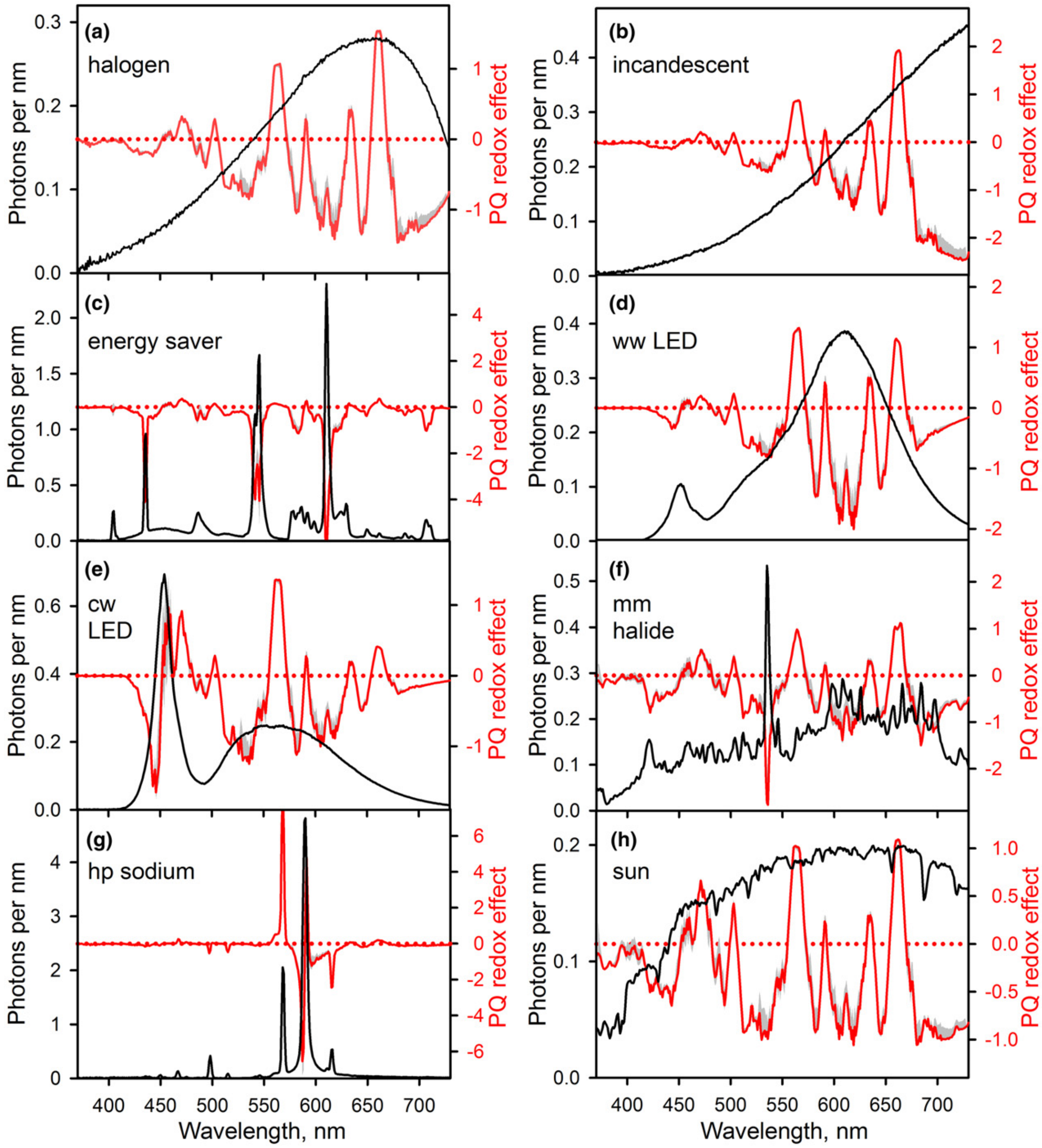

Figure 8. Contributions of individual wavelengths to the effects of different white light spectra on the redox state of the plastoquinone (PQ) pool.
Spectrum (black; left *y*-axis) and product of wavelength coefficient and spectrum (red; right *y*-axis) for KL-1500 halogen lamp (a), incandescent lamp (b), energy saver lamp (c), warm white LED (d), cool white LED (e), multimetal halide lamp (f), high-pressure sodium lamp (g) and sunlight (h). The spectrum of sunlight was measured on a cloudless day, April 14, in Turku, Finland. The spectra are normalized to equal photosynthetic photon flux density (PPFD). The dotted line is a zero level reference for the right *y*-axis. The gray shading shows the result of sensitivity analysis for 10% increase in χ^2.

state of the PQ pool in sunlight forms a light intensity measure (Figure 9). The PSI light nature of sunlight may explain why the redox state of the PQ pool is important for the acclimation to light intensity (Adamiec *et al*., 2008; Bode *et al*., 2016). Interestingly, most polychromatic light sources slightly favor PSI (Figures 8 and 9), and therefore plants can use PQ as a light intensity measure in artificial light, too.

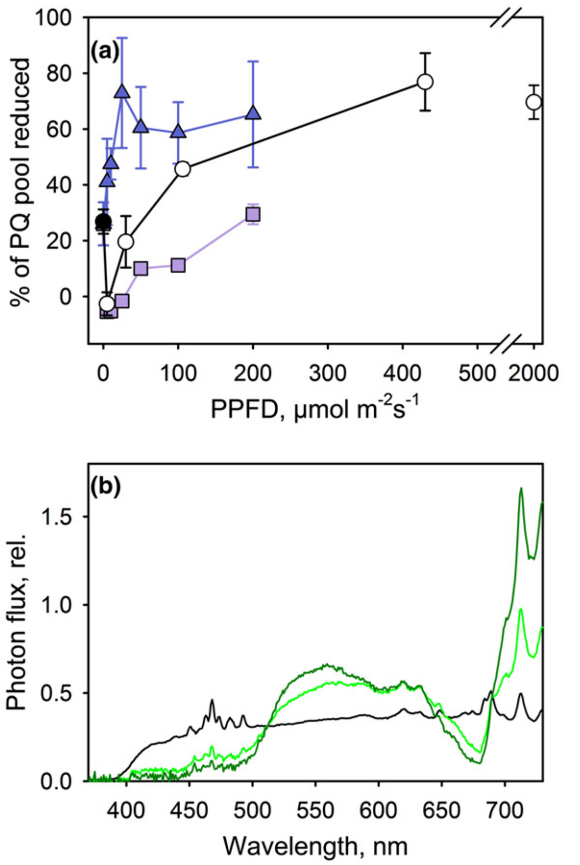

Figure 9. Effects of light intensity and wavelength distribution.
(a) The redox state of the plastoquinone (PQ) pool, measured from light-acclimated *Arabidopsis thaliana* leaves after 4 min in sunlight (circles), or in monochromatic Photosystem II (PSII) light of 470 nm (triangles) or monochromatic Photosystem I (PSI) light of 440 nm (squares) of different intensities. The solid circle shows the dark-acclimated state.
(b) Spectrum of a high-pressure Xenon lamp (black line) and the spectral effect of simulated canopy shade obtained by adding one (light green) or two overlapping leaves (dark green) of *Aponogeton ulvaceus*; photosynthetic photon flux density (PPFD) of each illumination experiment was adjusted to 50 μmol m^{-2} sec^{-1}. Each experimental point in (a) represents an average of three different leaves. The spectra in (b) were measured once.

Oxidation of the PQ pool in low-intensity sunlight (Figure 9a) suggests that low light intensity rather than enrichment of far-red light dominates the effect of canopy shade on the PQ pool. To test this, white light was filtered through thin leaves of an aquatic plant, *Aponogeton ulvaceus*. The spectrum of light filtered through the leaves showed the expected enrichment of far-red PSI light, but enrichment of 540–570 nm PSII light and suppression of blue PSI light (400–440 nm) would alleviate the effect of enrichment of far-red light (Figure 9b). Canopy shade probably induces far-red-specific phenomena mediated by cytoplasmic photoreceptors, but the data suggest that the light intensity effect of canopy shade might be regulated via redox state of the PQ pool. This suggestion is supported by the similarity of long-time acclimation responses under sunlight dimmed by filtering through canopy and under sunlight dimmed by neutral shade (Pons and de Jong-Van Berkel, 2004).

Relationship between PQ redox state, state transitions and phosphorylation of LHCII

Plant acclimation studies done so far have focused on illumination treatments that have been supposed to lead to an extreme State 1 or State 2. This approach does not tell whether a threshold fraction of the PQ pool must be reduced to activate the kinase and induce a transition to State 2, or whether intermediate reduction states of the PQ pool lead to stable intermediate states between States 1 and 2. To tackle this question, light-acclimated *A. thaliana* leaves were illuminated for 1 h using wavelengths with known effects on the redox state of the PQ pool, and the light state was deduced from 77 K emission spectra. In thylakoids isolated from leaves illuminated with 420, 520 or 690 nm PSI light, the ratio of PSI to PSII fluorescence (F735/F685) was 0.96 to 1.18, indicating that the leaves were in State 1, whereas illumination with PSII wavelengths (470, 560 or 660 nm) led to State 2 characterized with an F735/F685 of 1.35–1.44 (Figure 10). Light of 630 nm that has an intermediate effect on the PQ pool also caused an intermediate light state that was nearer to State 2 than State 1 (Figure 10a). The light state measured after 2-h dark incubation was also intermediate rather than extreme State 1 (Figure 10a). The phosphorylation level of LHCII was found to follow the redox state of the PQ pool and the light state of the plants (Figure 10b).

We found that the light state was curvilinearly related to the reduction of the PQ pool (Figure 10c). A fully oxidized PQ pool induced an extreme State 1, but State 2 was reached already at moderate reduction of the PQ pool, and was further deepened if the PQ pool was highly reduced. The results obtained after dark incubation fell on the same line (Figure 10c). This finding agrees with the activation of the STN7 kinase by binding of a single PQH_2 molecule to the Qo site of Cyt b_6f (Vener *et al*., 1997; Zito *et al*., 1999).

Various light regimes have earlier been used to induce State 1 or State 2 (Tullberg *et al*., 2000; Pesaresi *et al*., 2002; Piippo *et al*., 2006; Wagner *et al*., 2008; Shapiguzov *et al*., 2010). Application of our model to spectra of previous experiments (Figure S5) predicted that illumination with far-red light or with strongly far-red enriched red light fully oxidizes the PQ pool (Table S2), and that 470 nm light strongly reduces PQ (Table S2). However, the polychromatic broadband spectra used to induce State 2 (Tullberg *et al*., 2000; Pesaresi *et al*., 2002; Wagner *et al*., 2008) would reduce only 31.2–44.7% of the PQ pool (Table S2). This finding is in agreement with data in Figure 10(c) showing that already moderate reduction of the PQ pool induces State 2.

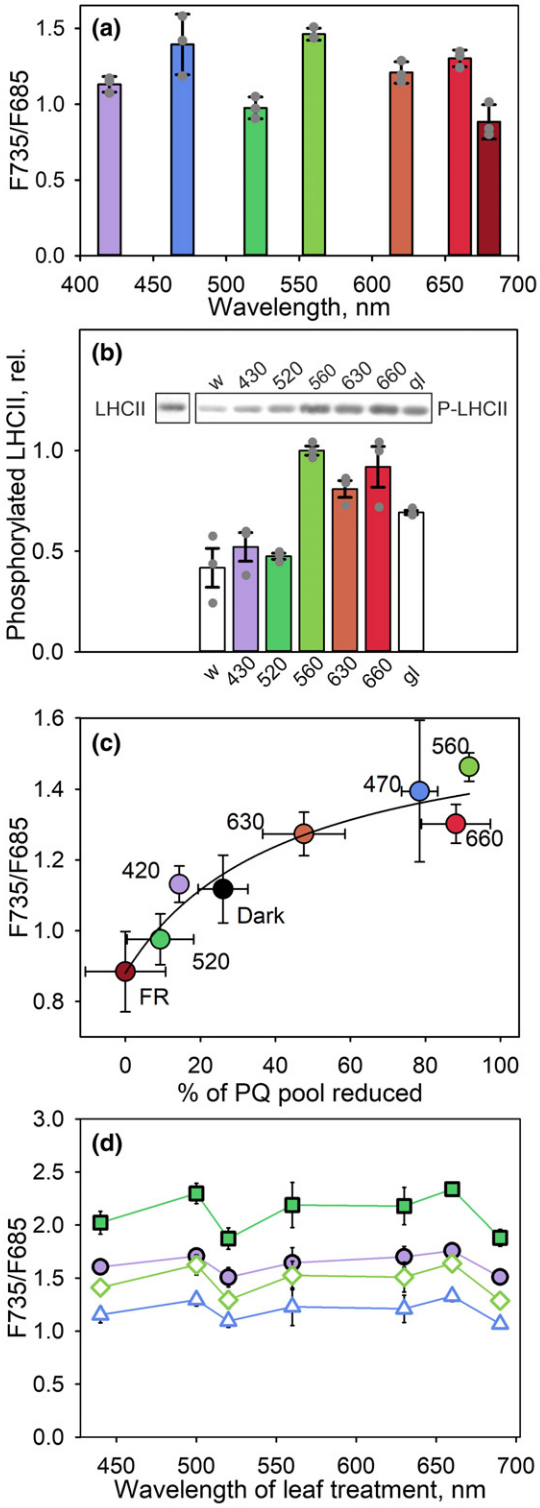

Figure 10. State transitions and phosphorylation of light-harvesting complex II (LHCII).
(a) 735/685 nm fluorescence ratio at 77 K in thylakoids isolated in the presence of NaF after 1-h illumination of *Arabidopsis thaliana* leaves at photosynthetic photon flux density (PPFD) 50 μmol m^{-2} sec^{-1} with monochromatic light of different wavelengths.
(b) Phosphorylation of LHCII after illumination of *A. thaliana* leaves with a halogen lamp (w, white bar) or with monochromatic light at indicated wavelengths (colored bars), and in leaves taken from a growth chamber (gl, white bar). A representative Western blot with a phosphothreonine antibody and identification of LHCII with a specific antibody in the same gel is shown. A sample containing 0.5 μg chlorophyll (Chl) was loaded to each well.
(c) Comparison of the redox state of the plastoquinone (PQ) pool, measured after 4-min illumination at PPFD 50 μmol m^{-2} sec^{-1}, and the 735/685 nm fluorescence ratio, measured after 1-h illumination at PPFD 50 μmol m^{-2} sec^{-1} with monochromatic light [colored circles; the PQ redox state data are from Figure 3 and the light state data from (a); the 630 nm fluorescence data were measured separately and the fluorescence from FR treatment after 680 nm illumination], or after 2 h in the dark (black circle); the numbers show the treatment wavelength. The line is a fit of all data to the descriptive hyperbolic equation $Y = 0.88 + 0.75X/(X + 43.5)$.
(d) The 735/685 nm fluorescence ratio measured from thylakoids isolated from *A. thaliana* leaves after 15-min illumination of the leaves at PPFD 50 μmol m^{-2} sec^{-1} with 440, 500, 520, 560, 630, 660 or 690 nm light. The actinic light in the fluorescence measurement was 440 [Photosystem I (PSI) light, solid circles], 470 [Photosystem II (PSII) light, open triangles], 520 (PSI light, solid squares) or 560 nm (PSII light, open diamonds). Each experimental point represents an average of measurements from three leaves, each from a different plant (a, b) or from thylakoids isolated from three different leaves, each from a different plant (b–d). The error bars show SEM, and the gray dots in (a) and (b) show results of individual experiments.

Visible light wavelengths that reduce or oxidize the PQ pool can tell about the mechanism of state transitions. For this, we brought *A. thaliana* leaves to State 1 (by illumination at 440, 520, 630 or 690 nm) or State 2 (illumination at 500, 560 or 660 nm), isolated thylakoids, and compared 77 K fluorescence spectra obtained with 440 (PSI), 470 (PSII), 520 (PSI) or 560 nm (PSII) excitation (Figure 10d; see Figure S6 for the spectra). The effect of state transition was quantified as the relative change in PSI/PSII fluorescence ratio from State 1 (after 690 nm illumination) to State 2 (after 660 nm illumination). The largest state transition effect, 27.3%, was obtained by using 560-nm PSII light for excitation, whereas 440-nm PSI light showed the smallest effect (16.4%). All PSII wavelengths produced a larger state transition effect than PSI wavelengths (Figure 10d). These data support a model in which LHCII, which is responsible for most light absorption of PSII, switches coupling between the photosystems during state transitions (Rochaix, 2014). In agreement with data in Figure 2(b), the excitation wavelength also affected the PSI/PSII fluorescence ratio (Figure 10d).

The significance of state transitions in nature is not completely clear. State transitions in higher plants balance the excitation of the two photosystems in low and moderate light (Rintamäki *et al.*, 1997), suggesting that they improve the efficiency of light use of leaves that are occasionally shaded by a canopy. An ability to adjust the excitation balance in a gradual manner may be important, as illumination may also vary gradually.

The redox state of PQ pool responds rapidly to light quantity and quality changes

The responses of the redox state of the PQ pool to changes in light intensity and quality were tested experimentally.

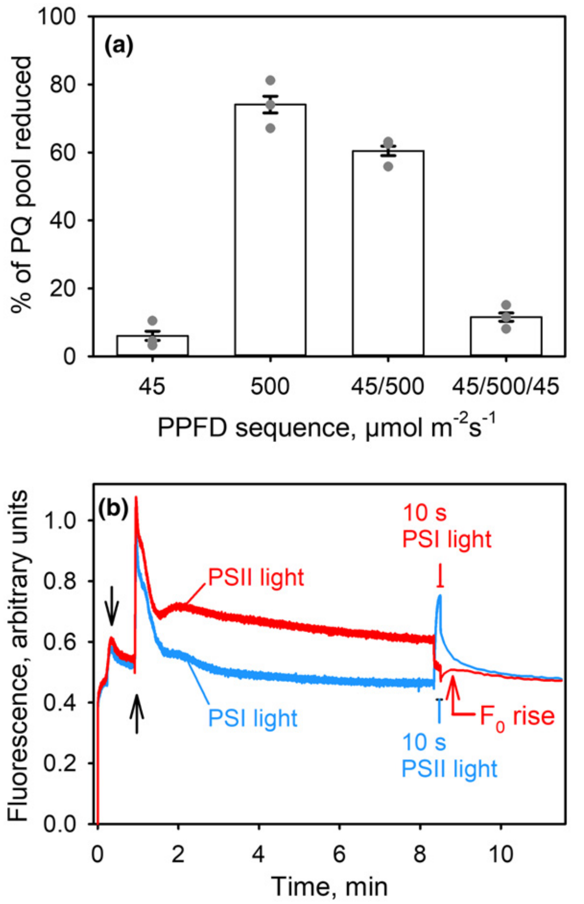

Figure 11. Response of the redox state of the plastoquinone (PQ) pool to fluctuations in light intensity and light quality.

(a) Redox state of the PQ pool measured after fluctuating light treatment [simulated sunlight, 4 min at photosynthetic photon flux density (PPFD) 45 μmol m^{-2} sec^{-1}, alternating with 1 min at 500 μmol m^{-2} sec^{-1}, as indicated]. Light-acclimated leaves were used for all illumination treatments in (a) and (b), except that the 100% reference value was obtained with a high light pulse on dark-acclimated leaves.

(b) Occurrence of F_0 rise after 10-sec pre-illumination with Photosystem I (PSI) light. An *A. thaliana* leaf was pre-illuminated at PPFD 50 μmol m^{-2} sec^{-1} for 8 min with green PSI light (520 nm), and then for 10 sec with green Photosystem II (PSII) light (560 nm; black line), or first with 560 nm light and then for 10 sec with 520 nm light (red line). After the second wavelength, pre-illumination was switched off and F_0 rise was probed using 660-nm activating light. The downward arrow indicates switch-on of activating light and the upward arrow indicates switch-on of PSII or PSI light; occurrence of F_0 rise is also shown with an arrow. Each bar in (a) and each curve in (b) represents an average of measurements from three different leaves [in (a) also from different plants], the error bars show SEM, and the gray dots in (a) show the results of individual measurements.

Fluctuations of light intensity, obtained with simulated sunlight, caused sequential reduction and oxidation of the PQ pool in a minute time scale (Figure 11). Fluorescence experiments were used to test the effect of light quality. For these experiments, we first asked how the quality of the pre-illumination light affects F_0 rise. For this, activated F_0 rise was measured from *A. thaliana* leaves with monochromatic light both as pre-illumination and activating light. These experiments revealed that F_0 rise only occurs if the pre-illumination light oxidizes the PQ pool (Figure S7). For testing how rapidly the PQ pool can change its redox state, an *A. thaliana* leaf was first pre-illuminated at PPFD 50 μmol m^{-2} sec^{-1} with green PSI light (520 nm) to oxidize the PQ pool, but the wavelength was changed for the last 10 sec of the pre-illumination to green PSII light (560 nm). Finally, weak activating light was applied to measure F_0 rise. No F_0 rise was seen with this illumination protocol (Figure 11b). On the other hand, if 560 nm PSII light was applied first and 520 nm PSI light was switched on for the last 10 sec, then an F_0 rise occurred (Figure 11b). These data show that the oxidation state of the PQ pool changed in the light within 10 sec. A simple calculation confirms that a small difference in the absorption rates of the photosystems leads to oxidation or reduction of the PQ pool in a few seconds. For example, assuming 48.5% mean absorptance of PSII, an average ETR of 13.5 μmol electrons m^{-2} sec^{-1} (Figure 5) and 2.41 μmol photochemically active PQ m^{-2}, a 5% difference in the ETR of PSII and PSI, would reduce or oxidize the PQ pool within 3.4 sec.

CONCLUSIONS AND FUTURE PROSPECTS

The present results show that wavelengths favoring one of the photosystems over the other occur throughout the visible range, and the effect of white light of moderate intensity depends on its spectrum. Most forms of white light, including natural sunlight, appear to favor PSI. This is important for the regulatory functions of the PQ pool, as light intensity has a gradual effect on the redox state of the PQ pool only under PSI light, and therefore the PQ pool can actually measure the intensity of sunlight. The results were used to construct a model for the prediction of the redox state of the PQ pool under polychromatic light of any type. The finding that the redox state of the PQ pool can be almost freely adjusted with moderate photosynthetically active light opens new possibilities for physiological studies concerning the role of PQH_2 as a producer and scavenger of reactive oxygen species (Kruk and Trebst, 2008; Khorobrykh and Tyystjärvi, 2018). Plant biotechnology may also benefit from the possibility to control a key signaling factor by the spectral distribution of plant growth light.

The finding that a gradual change in the light state correlates with a gradual change in the redox state of the PQ pool confirms that the light state depends on the redox state of the PQ pool. The gradual effect is also important for the mechanism of the regulation of state transitions. The elucidation of the action spectrum of the redox state of the PQ pool allows testing the role of the PQ pool in state transitions, gene regulation and other acclimation reactions under photosynthetically active light without using photosynthetic inhibitors like DCMU and DBMIB. Furthermore, several wavelengths can be compared in signaling studies to avoid ambiguity between regulation by specific photoreceptors and regulation by the PQ pool, and the availability of photosynthetically active wavelengths favoring PSII or PSI makes it possible to study light intensity and light quality responses independently of each other.

EXPERIMENTAL PROCEDURES

Plant material

Wild-type (accession Col-0) and the *ndhO* mutant (Rumeau *et al.*, 2005) of *A. thaliana* L. were grown for 5–6 weeks at 8 h light/16 h dark, PPFD 150 μmol m^{-2} sec^{-1} from Powerstar HQI-BT lamps (Osram, Munich, Germany). Tobacco (*N. tabacum* L.) and the *hcef* mutant of *A. thaliana* (Livingston *et al.*, 2010) were grown for 2 months at 16 h light/8 h dark, PPFD 150 μmol m^{-2} sec^{-1}.

Measurements of Chl *a* fluorescence, P_{700}, PC kinetics and absorptance from intact leaves

Pulse amplitude modulated Chl *a* fluorescence was measured with a PAM-101 fluorometer (Heinz Walz GmbH, Effeltrich, Germany) modified to allow computer control of the MB (Tyystjärvi and Karunen, 1990). Unless otherwise indicated, activated F_0 rise was measured as follows. An *A. thaliana* leaf or a piece of a *N. tabacum* leaf was cut from a light-acclimated plant, brought on moist paper under the fluorometer's probe and kept in the dark for 2 min. Thereafter, the MB (setting 3) was switched on and, after a few seconds, monochromatic activating light, PPFD 2.5 μmol m^{-2-} sec^{-1}, defined with a 10 nm HWHM Corion line filter (Newport, Irvine, CA, USA), was switched on. After 1 min, a saturating pulse was fired and pre-illumination light (47.5 μmol m^{-2} s^{-1}) was switched on. Pre-illumination light and activating light summed up to PPFD 50 μmol m^{-2} sec^{-1}. After 7.5 min, a second saturating pulse was fired to make sure that NPQ was not significantly changed during the pre-illumination. After one more minute of illumination, actinic light and the MB were switched off, but the activating light remained on. Thereafter, the MB was used in cycles of 0.6 sec on/5 sec off. PPFD was measured with a quantum sensor (Li-Cor, NE, USA) calibrated for measurements with monochromatic light. A KL-1500 illuminator (Heinz Walz GmbH) was used as a white light source. *Nicotiana tabacum* was chosen for the action spectrum of F_0 rise because *N. tabacum* leaves are large and uniform, and the attached leaves stay in a similar physiological state for a long time, and therefore it was easy to get physiologically homogenous material for a long measurement series.

For quantification of F_0 rise, fluorescence data were divided by the steady fluorescence level obtained before switching off the pre-illumination. The area limited by the fluorescence curve and the fluorescence level at the first F_0 data point, measured 5 sec after the pre-illumination, was integrated from 5 to 80 sec. Fluorescence values smaller than the 5 sec data point contributed negatively.

Φ_{II}, ETR (Genty *et al.*, 1989) and qL (Kramer *et al.*, 2004) were measured with a PAM-2000 fluorometer (Heinz Walz GmbH) from *A. thaliana* leaves by illuminating a light-acclimated leaf at each wavelength through a Corion line filter (PPFD 50 μmol m^{-2} sec^{-1}, 4 min) and then firing a saturating flash (PPFD 2000 μmol m^{-2} sec^{-1}, 0.8 ms). ETR was calculated with the equation PPFD $\times \alpha(\lambda) \times \Phi_{II}(\lambda) \times 0.485$, where $\alpha(\lambda)$ and $\Phi_{II}(\lambda)$ are the leaf absorptance and quantum yield of PSII electron transfer as functions of wavelength (λ). The factor 0.485 is the PSII/(PSII + PSI) ratio calculated from the PSI/PSII ratio of 1.06, as earlier measured for the *A. thaliana* accession Col-0 grown in similar conditions (Yin *et al.*, 2012). ETR was not calculated for wavelengths below 420 nm where a large deviation of the F_0 yield from the leaf absorptance indicated significant absorption by non-photosynthetic compounds. The dark-acclimated value of maximum fluorescence was measured from the same leaf. Leaf absorptance values at different wavelengths were measured from sets of 10 *A. thaliana* leaves using an integrating sphere (Labsphere, North Sutton, NH, USA) with a calibrated STS-VIS spectrometer (Ocean Optics Duiven, The Netherlands). Matt black cardboard with a calibrated reflection spectrum (~2% reflection throughout the visible range) was used to calibrate the absorptance of the sphere.

A combination of the P_{700}^{+} absorbance signal and the PC^{+} signal (Schreiber *et al.*, 1988; Schreiber and Klughammer, 2016) was measured at 830 nm with an EDP700DW accessory of PAM-101. Monochromatic illumination with visible light wavelengths was provided through a second branch of the light guide using a Corion line filter to define the wavelength.

Light-induced changes in the redox state of PC and P_{700} were measured from *A. thaliana* leaves with the Dual/KLAS-NIR spectrophotometer (Heinz Walz GmbH). The spectrometer measures the redox changes of PC and P_{700} through deconvolution of modulated absorbance spectra. The light guide of the instrument was modified to allow illumination of the leaf sample from the adaxial side with an arbitrary light source. In each experiment, a light-acclimated *A. thaliana* leaf was first kept in the instrument in the dark for approximately 1 min, during which the instrument automatically adjusts a baseline for the measurement. Changes in the redox states of P_{700} and PC were then followed during 4 min of illumination with monochromatic light (PPFD 50 μmol m^{-2} sec^{-1}). In a separate experiment, the external illumination was given through a 1-msec electronic shutter (Uniblitz; Vincent Associates, Rochester, NY, USA) to probe the fast kinetics of re-reduction of PC and P_{700}. The maximum signals of the far-red-oxidizable P_{700}^{+} and PC were measured after the illumination. A 30-sec darkness after illumination was followed by switch-on of a 735-nm far-red light of the instrument and a saturating pulse on top of the far-red light (Schreiber and Klughammer, 2016).

The action spectrum of initial fluorescence F_0 was measured by illuminating a dark-acclimated *A. thaliana* leaf with a 1000 W high-pressure Xenon illuminator (Sciencetech, London, ON, Canada) through a Corion line filter, a 750-nm short-pass filter (FES0570; Thorlabs, Newton, NJ, USA), a liquid light guide (Series 300, Lumatec GmbH, Oberhaching, Germany) and a bifurcated 3-mm plastic light guide (Edmund Optics, Barrington, NJ, USA). A 700-nm Corion short-pass filter LS-700-S was additionally used in the 430–700 nm range. Fluorescence was detected with a photomultiplier tube through a 750-nm Corion line filter. Relative fluorescence yield was calculated by dividing the measured fluorescence intensity with PFD measured through a second branch of the plastic light guide with an STS-VIS spectrometer (Ocean Optics). To prevent actinic effects of the illumination, an electronic shutter (Vincent Associates) was used to provide light only for 0.1 sec every 5 sec. Each data point was calculated as an average of five readings.

Chlorophyll *a* fluorescence

For measurements of the excitation spectrum of the PSII/PSI fluorescence ratio, thylakoids were isolated from two–three *A. thaliana* leaves after an overnight dark period as earlier described (Hakala *et al.*, 2005) and suspended at 10 μg Chl ml^{-1} in a buffer containing 0.5 M sorbitol, 5 mM NaCl, 10 mM $MgCl_2$, 10 mM Hepes-KOH, pH 7.4. Spectra were measured at 77 K with a QEPro spectrometer (Ocean Optics) from 50-μl thylakoid samples. Comparison of the spectral response of the spectrometer with the emission spectrum of a calibration lamp LS-1-CAL-220 (Ocean Optics) indicated a flat response at the 680–740 nm range. Excitation light was obtained from a slide projector through the Corion line filters. The PSII/PSI fluorescence ratio was defined as the ratio of the PSII peak at 685 nm to the PSI peak at 735 nm.

State transitions were measured by illuminating an *A. thaliana* leaf through a Corion line filter (PPFD 50 μmol m^{-2} sec^{-1}) for 15 min or 1 h, as indicated, and isolating thylakoids, supplementing all buffers with 10 mM NaF to preserve the phosphorylation level during isolation (Rintamäki *et al.*, 1997). Emission spectra were measured as described above with excitation at 440, 470, 520 or 560 nm, as indicated. The state transition effect on 77 K fluorescence was calculated as $((F735/F685)_{State\ 2} - (F735/F685)_{State\ 1})/(F735/F685)_{State\ 1}$, where State 2 was induced by illuminating the leaves, prior to isolation of thylakoids, for 15 min with 660 nm light, and State 1 was obtained by illuminating with 690 nm light.

Fluorescence induction curves were measured with PAM-101 fluorometer (Heinz Walz GmbH) from *A. thaliana* leaves in red light, PPFD 80 μmol m^{-2} sec^{-1}. Before the measurements, each leaf was vacuum infiltrated with buffer solution (0.3 M sorbitol, 5 mM NaCl, 10 mM $MgCl_2$, 10 mM Hepes-KOH, pH 7.4, 2.5% ethanol) in the presence or absence, as indicated, of 50 μM DCMU and dark incubated for 2 h in a drop of the respective buffer solution. The CA above the fluorescence induction curve, from the beginning of actinic illumination to time point corresponding to F_M, was normalized by dividing by F_M. The number of photochemically active PQ molecules per PSII was calculated as $0.5 \times (CA_{NO\ ADDITION} - CA_{DCMU})/CA_{DCMU}$.

HPLC measurements

For HPLC measurements, *A. thaliana* leaves were illuminated at a number of conditions as specified below. A 1000 W high-pressure Xenon lamp (Sciencetech) equipped with a 9-cm water filter was used as a light source, unless otherwise stated. White light was obtained through an ultraviolet protection film (Long Life for Art, Eichstetten, Germany), and the same Corion line filters as used for F_0 rise measurements were used to obtain monochromatic light. Additional measurements in monochromatic light were done using a 3 W 380-nm LED equipped with a Corion line filter, FWHM 10 nm, with 420, 471, 519, 635, 660 and 720 nm LEDs (3 W; FWHM of emission 17, 10, 12, 16, 17 and 30 nm, respectively), and with a 3 W cool white LED, filtered through a 554, 36 nm FWHM filter (LF 101537, 560/40; Delta Optical Thin Film A/S, Hørsholm, Denmark). The 471 and 519 nm LEDs were equipped with the Newport 10BPF10-470 and the Delta Optical Thin Film LF 101724, 522/12 filter, respectively. LEDs were bought from LED Fedy, Shenzen, People's Republic of China, unless otherwise stated. For testing polychromatic illumination, experiments were done with a KL-1500 halogen lamp (Heinz Walz GmbH), 60 W incandescent lamp (Airam Electric, Kerava, Finland), a 12 W energy saver lamp (Superstar, Osram), a 3 W cool white LED and a 6 W E27 warm white LED lamp Pirkka E27 (Kesko, Helsinki, Finland), a 400 W multimetal halide lamp and a 400 W high-pressure sodium lamp, as indicated. Experiments in natural sunlight were done by exposing a light-acclimated leaf to sunlight for 4 min in cloudless summer day conditions in a temperature-controlled box at 20°C. Glossy steel nets, tested for negligible effect on the spectrum of sunlight, were used to reduce the PPFD of sunlight to predefined values.

For testing the effect of simulated canopy shade, light from a high-pressure Xenon lamp was filtered through the leaves of *A. ulvaceus* obtained from an aquarium shop. A LED-based sunlight simulator module (SLHolland, Breda, The Netherlands) was used to test the effect of fluctuating light on the redox state of the PQ pool.

The measurement of the redox state of the PQ pool was based on a published method (Kruk and Karpinski, 2006). For each measurement, a leaf was excised from a light-acclimated *A. thaliana* plant and illuminated with the treatment light for 4 min at PPFD 50 μmol m^{-2} sec^{-1}, unless otherwise stated. To obtain a fully reduced photochemically active PQ pool for reference, an *A. thaliana* leaf was dark acclimated on moist paper for 2 h and illuminated at 2000 μmol m^{-2} sec^{-1} of white light for 30 sec. After the treatment, the leaf was ground in a mortar for 10–20 sec in 1 ml of cold ethyl acetate that was pre-dried with molecular sieves, 4 Å (Sigma Aldrich, St Louis, MO, USA). The illumination treatment was continued throughout the grinding of the leaf. A transparent glass mortar, illuminated through the bottom, was used to test if shadowing by the pestle compromises the continuation of the illumination throughout the grinding of the leaf.

After grinding, the extract was filtered through a 0.2-μm polytetrafluoroethylene filter, evaporated to dryness under a nitrogen flow, dissolved to methanol:hexane (34:2, v:v), and divided into two parts. One part was immediately assayed with HPLC, and 0.5 μl of fresh 200 mM $NaBH_4$ in methanol was added to 50 μl of the other part to reduce all PQ to PQH_2 before assaying with HPLC. The peaks of PQ and PQH_2 were identified on the basis of retention time (Kruk and Karpinski, 2006) and the behavior by reduction with $NaBH_4$; PQ was also confirmed by its absorption spectrum. The percentage of PQH_2 in the photochemically active PQ pool was calculated as $100 \times [R_S - R_{690}]/[R_{HL} - R_{690}]$, where each R represents the HPLC peak area of PQH_2 divided by the peak area of PQH_2 in the same sample after reduction with $NaBH_4$. The subscript S refers to the sample under investigation, and subscripts 690 and HL refer to the fraction of PQH_2 obtained for leaves illuminated for 4 min with 690 nm light (50 μmol m^{-2} sec^{-1}), and for leaves illuminated with high light after dark acclimation, respectively. The preparation of the PQ standard, used for the quantification of PQ, has been described elsewhere (Khorobrykh *et al.*, 2020).

The ratio of PQ to Chl was measured from batches of three *A. thaliana* leaves from which PQ and Chl were extracted with MeOH, and the extract was used for measurement of both Chl and PQ. Chl was measured by absorption (Porra *et al.*, 1989). PQ was estimated with HPLC (Kruk and Karpinski, 2006). Briefly, PQ was reduced to PQH_2 by $NaBH_4$ and applied in HPLC, and the area of the PQH_2 peak was measured. The ratio between area of peak and concentration of PQH_2 was estimated using the PQ standard.

Phosphorylation of LHCII

For measurements of the amount of phosphorylated LHCII, *A. thaliana* leaves were illuminated for 8 min with 430, 520, 560, 630 or 660 nm light or with white light from a KL-1500 illuminator (PPFD 50 μmol m^{-2} sec^{-1}), or taken directly from the growth chamber. Thylakoids were isolated and proteins from samples containing 0.5 μg Chl were separated and transferred to a membrane as earlier described (Rintamäki *et al.*, 1997). The membrane was blocked with 5% bovine serum albumin, and antibodies against LHCII (Lhcb1, Agrisera, Vännäs, Sweden) and phospho-Thr (New England Biolabs, Ipswich, MA, USA) were used for detection. A goat anti-rabbit IgG (H + L) alkaline phosphatase conjugate (Zymed, Thermo Fisher Scientific, Waltham, MA, USA) and the CDP Star chemiluminescence kit (New England Biolabs) were used for detection in Western blotting. The immunoblots were quantified with a FluorChemi image analyzer (Alpha Innotech, Land Leandro, CA, USA).

Model for the redox state of the PQ pool in polychromatic light

The redox state of the PQ pool obtained by illuminating a leaf with spectrum i (i = 1, 2, ..., 25) was calculated as a dot product $\mathbf{s}_i \bullet \mathbf{c}$, where each $\mathbf{s}_i$ is a vector containing the contribution of each wavelength λ (λ = 370, 371, … ,730) in spectrum i and each element of vector $\mathbf{c}$ in which each element $\mathbf{c}_\lambda$ is the coefficient for wavelength λ (λ = 370, 371, … ,730). The error of the result was calculated as the square of the norm of the vector $\mathbf{Sc}$-$\mathbf{p}$, where $\mathbf{S}$ is a matrix consisting of $\mathbf{s}_i$ as the ith line, and $\mathbf{p}_i$ is the experimentally determined percentage of PQH_2 in the PQ pool under illumination with a light source emitting spectrum i. The spectrum of each light source was normalized to the same PPFD by dividing the contribution of each wavelength from 370 to 730 nm by the sum of the contributions of wavelengths from 400 to 700 nm.

The coefficient vector $\mathbf{c}$ was obtained by minimizing the sum

$$|\mathbf{Sc} - \mathbf{p}|^2$$

by allowing each $\mathbf{c}_\lambda$ to run free in a Monte Carlo procedure (8×10^7 rounds) with simulated annealing (cooling by a factor of $0.004 \times \mathbf{c}_\lambda$ after each round that led to decrease of the error and heating by a factor of $0.6 \times \mathbf{c}_\lambda$ at every 10th round that led to decrease of the error). To add flexibility to the model, the maximum absolute value of a coefficient was set to 55, which allows the theoretical maximum amount of oxidized or reduced PQ to exceed the total amount by 5% at each wavelength. The maximum difference between two consecutive coefficients was set to 10 to avoid overfitting.

Sensitivity analysis was done after fitting by finding both a positive and negative increment that, when added to coefficients $\mathbf{c}_\lambda \ldots \mathbf{c}_{\lambda+4}$ (λ = 370, ..., 726) led to 10% increase in $|\mathbf{Sc} - \mathbf{p}|^2$, and the sensitivity estimates were calculated by separately averaging the positive and negative increments for each wavelength (374–726 nm).

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AUTHOR CONTRIBUTIONS

ET, HM and SK designed and planned the research. HM, SK, MH-Y, VH, IK and TA performed the experiments. ET, HM, SK, MH-Y, VH and TT evaluated the data. HM, SK, TT and ET wrote the manuscript with contributions from all authors. All authors reviewed, revised and approved the article for publication.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Original data pertinent to the article are deposited in Mendeley at https://doi.org/10.17632/5grmz2gv8b.1. The software designed to calculate the wavelength coefficients is available at http://github.com/esatyy/Wavelength-coefficients.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. Fluorescence induction curves of vacuum-infiltrated *Arabidopsis thaliana* leaves in the absence (a) and presence (b) of DCMU. Fluorescence induction was measured with the PAM-101 fluorometer after vacuum infiltration and 2 h of dark incubation. Both curves represent average curves measured from six leaves from three different plants. The CA above the curve and below the maximum of the curve was calculated to measure the ratio of photochemically active PQ to PSII.

Figure S2. Reduction of PQ by $NaBH_4$. (a, b) A HPLC chromatogram of pure PQ in methanol and (c, d) a chromatogram of a leaf extract obtained after 4 min illumination of the leaf in 690-nm light. Chromatograms before (black line) and after (red line) addition of $NaBH_4$, showing the absorbance at 255 nm (a, c) and the fluorescence at 330 nm (b, d). The retention time of PQ is 25 min (a, c) and that of PQH_2 is 7.5 min (b, d).

Figure S3. Relationship between closure of PSII centers (1-qL) and reduction state of the PQ pool. *Arabidopsis thaliana* leaves were illuminated with monochromatic light, and the reduction state of the PQ pool was measured with HPLC. The 1-qL parameter was measured during illumination with the same wavelengths and the same PPFD, 50 μmol m^{-2} sec^{-1}. The PQ data are from Figure 3, and the 1-qL data are from Figure 5. The numbers show the wavelength in nm of each treatment.

Figure S4. Kinetics of redox changes of PC and P_{700} after illumination with monochromatic light. Light-acclimated *Arabidopsis thaliana* leaves were illuminated for 4 min at PPFD 50 μmol m^{-2} sec^{-1} with 440 (a), 520 (b), 630 (c), 690 (d), 470 (e), 560 (f) or 660 nm light (g). The arrows show when the illumination was ceased by closing a 1-msec shutter. Redox changes were measured with a Dual/KLAS-NIR spectrometer. The maximum refers to the maximum signal, measured during a saturating flash after far-red pre-illumination. Each picture depicts a single, representative measurement out of three measurements, each from a different plant.

Figure S5. Spectra of PSII and PSI lights from the literature. (a) Spectrum of PSI light used by Wagner *et al*. (2008) (black solid line, redrawn from the source), Tullberg *et al*. (2000) (red dashed line, converted to a photon number spectrum from the published spectrum) and Piippo *et al*. (2006) (blue dotted line, converted to a photon number spectrum from the published spectrum). (b) Spectrum of PSII light used by Wagner *et al*. (2008) (black solid line, redrawn from the source), Pesaresi *et al*. (2002) (green dotted line; the spectrum was reconstructed by multiplying the measured emission spectrum of the KL-1500 illuminator and the transmission spectrum of the BG39 filter, obtained from the manufacturer), and by Tullberg *et al*. (2000) and Piippo *et al*. (2006) [red dashed line; the same PSII light was used in both papers, and the spectrum obtained from Piippo *et al*. (2006) was converted to a photon number spectrum].

Figure S6. Fluorescence spectra. Arabidopsis leaves were illuminated for 15 min at 520 (PSI light, solid line), 560 (PSII light, dotted line), 660 (PSII light, dash-dotted line) or 690 nm (PSI light, dashed line), thylakoids were isolated in the presence of NaF and emission spectra were measured at 77 K, using 440-nm PSI light (a) or 470-nm PSII light (b) for excitation. The spectra have been normalized to the 685-nm value. Each spectrum represents an average of spectra measured from three thylakoid batches, each isolated from an individual treated Arabidopsis leaf.

Figure S7. Combined action spectrum of pre-illumination and activating light in measurements of activated F_0 rise. (a) Examples of activated F_0 rise curves obtained with selected wavelength pairs. The first number refers to the wavelength of the pre-illumination and the second to that of the activating light, and the designations PSII and PSI indicate which photosystem is favored. (b) Quantification of F_0 rise obtained with selected wavelength pairs. In all experiments, pre-illumination was 8 min at PPFD 50 μmol m^{-2} sec^{-1} and the PPFD of the activating light was 2.5 μmol m^{-2} sec^{-1}. Each measurement was done from an individual Arabidopsis leaf.

Table S1. Validation of the use of a ceramic mortar for measurement of the redox state of the PQ

Table S2. Predicted percentage of PQH_2 during illumination with light sources known to lead to formation of State 1 or State 2

Appendix S1. The coefficients shown in Figure 7(b) in table form.

Open Research Badges

This article has earned Open Data and Open Materials badges. Data and materials are available at https://www.ncbi.nlm.nih.gov/bioproject/436616 and https://github.com/smoenga55/Moenga-2020/blob/master/Moenga_2020_Git.Rmd.

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